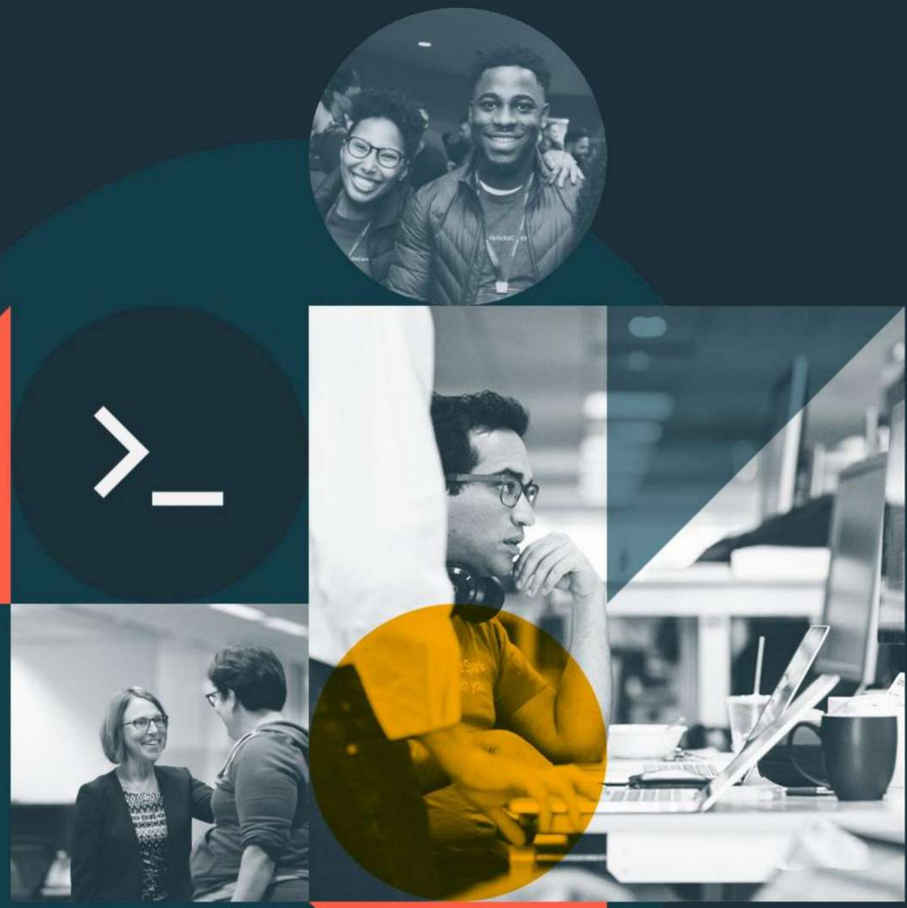
Manage Data with Delta Lake



Module 03

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Module Agenda

Manage Data with Delta Lake

What is Delta Lake

DE 3.1 - Schemas and Tables

DE 3.2 - Version and Optimize Delta Tables

DE 3.3L - Manipulate Delta Tables Lab

DE 3.4 - Set Up Delta Tables

DE 3.5 - Load Data into Delta Lake

DE 3.6L - Load Data Lab

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What is Delta Lake?

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Delta Lake is an open-source project that enables building a data lakehouse on top of existing cloud storage

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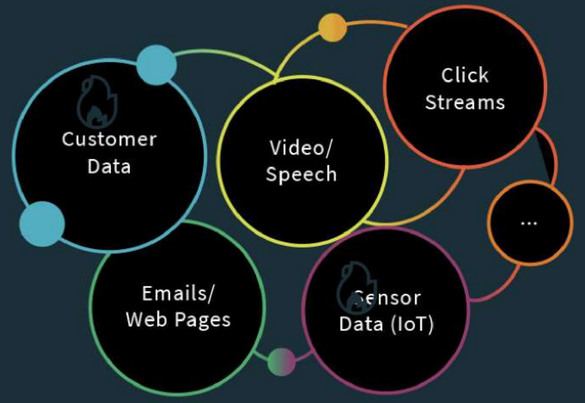
The Promise of the Data Lake

1. Collect Everything



Garbage In

2. Store it all in the Data Lake

Garbage Stored

3. Data Science & Machine Learning

- Recommendation Engines
- Risk, Fraud Detection
- IoT & Predictive Maintenance
- Genomics & DNA Sequencing

Garbage Out

What Delta Lake is ?

- Delta Lake is an open-source data management platform.
- It enhances data operations and analytics.
- It leverages standard data formats.
- Optimized for cloud object storage.
- Built for efficient metadata handling.

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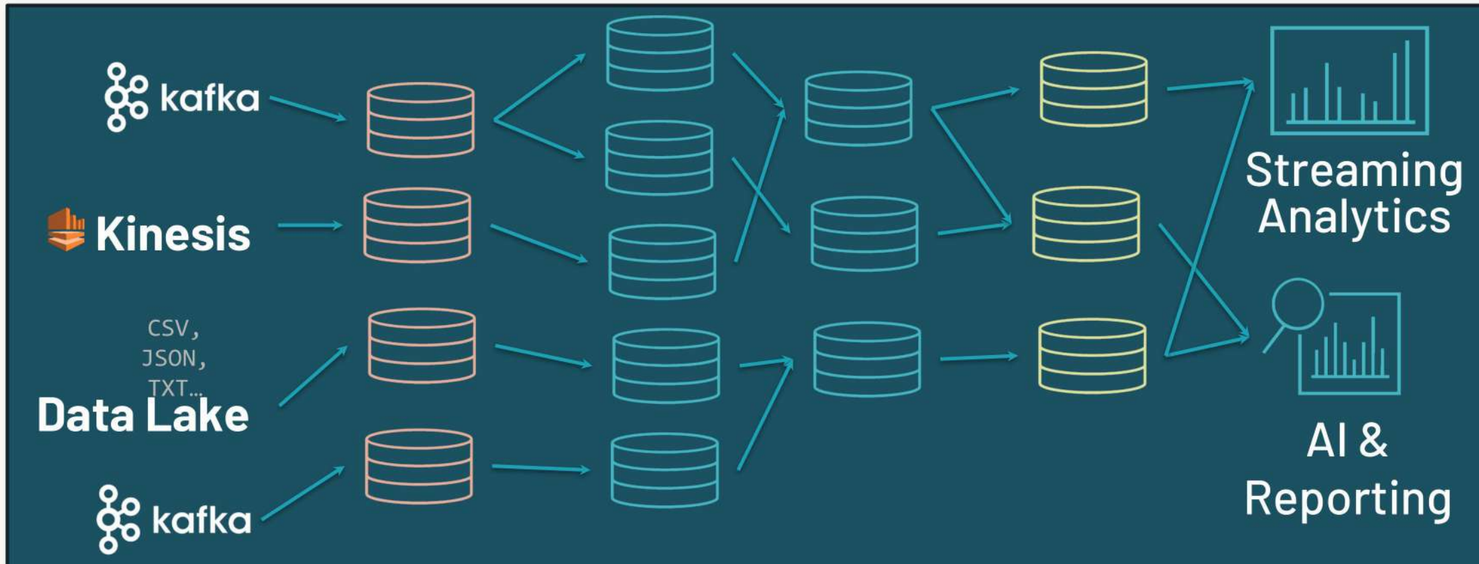
The Delta Lake Architecture



Full ACID Transaction

Focus on your data flow, instead of worrying about failures.

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Problems solved by ACID

- **Streamlined Data Append :** Delta Lake simplifies and improves data appending, making it efficient even with concurrent writes.
- **Simplified Data Modification :** Delta Lake simplifies data modification, ensuring data consistency.
- **Data Integrity Through Job Failures :** Delta Lake prevents data inconsistencies due to job failures, maintaining data integrity.
- **Support for Real-time Operations :** Delta Lake serves as a robust data source and sink for real-time and streaming operations.
- **Efficient Historical Data Version Management :** Delta Lake offers time travel for accessing historical data versions, and its cost-effectiveness depends on your specific use case and alternative solutions.

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Key Features of Delta Lake

- **Update and Delete:** Delta Lake allows for the modification and removal of records, offering a crucial distinction from other data formats.

- **Data Skipping Index:** Delta Lake employs file statistics to optimize query performance by skipping unnecessary data scans.

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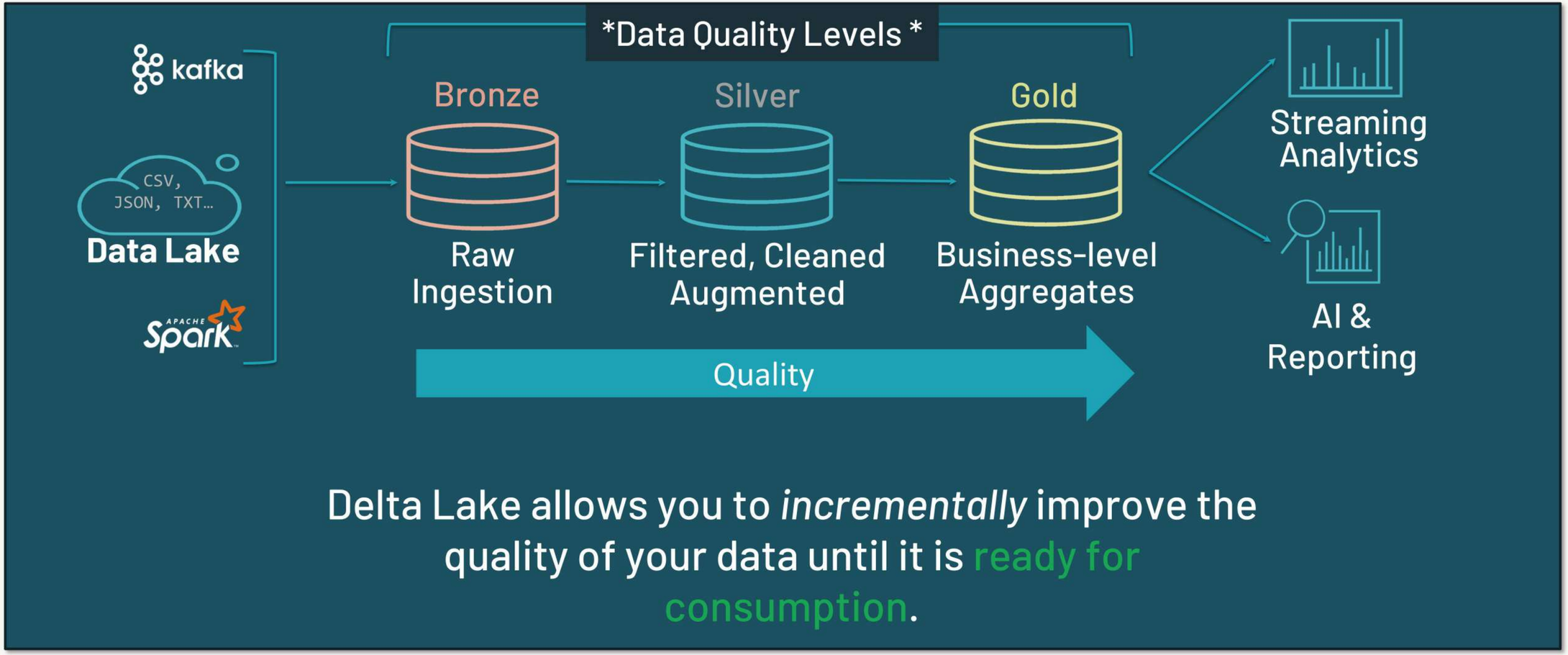
The Delta Lake



Delta Lake allows you to *incrementally* improve the quality of your data until it is ready for consumption.

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Delta Lake is the default format for tables created in Databricks

```
CREATE TABLE foo
USING DELTA
```

```
df.write
  .format("delta")
```

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DE 3.1: Schemas and Tables

Create schema (database) as repository for your tables/views

Create managed and external Delta tables

Insert records in Delta Lake tables

Dropping Delta Lake tables

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DE 3.2: Version and Optimize Delta Tables

Highlight that OPTIMIZE improves file size management for better performance.

Emphasize that ZORDER is a method to organize data efficiently within files.

Introduce Data Skipping as a crucial feature for optimizing query performance.

Explain the directory structure of Delta Lake files.

Describe how Delta Lake maintains a transaction history for your tables.

Mention the capability to query and roll back to previous table versions.

Discuss the VACUUM and DRY RUN operations for removing unnecessary data files

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DE 3.3L - Manipulate Delta Tables Lab

Review table history

Query previous table versions and rollback a table to a specific version

Perform file compaction and Z-order indexing

Preview files marked for permanent deletion and commit these deletes

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DE 3.4: Set up Delta Tables

Using CTAS statements to create Delta Lake tables

Creating new tables from existing views or tables

Declaring table schema with generated columns and descriptive Comments

Setting options for location, constraints, and partitions

Creating deep and shallow clones

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DE 3.5: Load Data into Delta Tables

CREATE OR REPLACE TABLE

Overwrite data tables using INSERT OVERWRITE

Append to a table using INSERT INTO

Append, update, and delete using MERGE INTO

Ingest data incrementally into tables using COPY INTO

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DE 3.6L: Load Data Lab

Create an empty Delta table with a provided schema

INSERT INTO from an existing JSON table into a Delta table

Use a CTAS statement to create a Delta table from files

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